ELTAI501 – ELECTRICAL AUTOMATION SYSTEM INSTALLATION

DATE: .../.../2025
PERIOD: .....

INSTITUT TECHNIQUE BAPTISTE (ITB TSS)
Study Success Discipline

TECHNICAL SECONDARY SCHOOLS (TSS)

END OF TERM I EXAMINATIONS: 2025-2026

QUESTION PAPER

SECTOR: Energy

TRADE NAME: Electrical Technology

RQF LEVEL: 5

MODULE TITLE: ELECTRICAL AUTOMATION SYSTEM INSTALLATION

MARKS: ...../100

DURATION: 3 HOURS

INSTRUCTIONS TO CANDIDATES:

This Exam paper is composed of Twenty-four (24) questions distributed in Three Sections (A, B, C). Follow the instructions given below, and answer the indicated questions in each section for a total of 100marks

Section **A**: Between (11-17) questions, all are **Compulsory**. **55 marks**
Section **B**: Five (5) questions, choose and answer any three (3). **30 marks**
Section **C**: Two (2) questions, choose and answer any one (1) **15 marks**

Allowed Materials: Blue pen, Ruler, Calculator, Pencil, (Fitting with the exam paper)

SECTION A: Attempt all questions **(55 marks)**

01.	**1. Which of the following is the primary advantage of integrating a Distributed Control System (DCS) with an automation system?** A) Reduced power consumption B) Centralized control with high redundancy C) Decentralized control with high system reliability D) Reduced system complexity and lower maintenance cost	**(2Marks)**
02.	**2. When designing the control circuit for an automation system, which of the following is the most critical factor for ensuring reliable communication between components?** A) Control signal power supply consistency B) Proper configuration of communication protocols like Modbus RTU or TCP/IP C) Frequency of data polling D) The physical distance between PLC and I/O modules	**(2Marks)**
03.	**3. In an automation system, which of the following best describes the function of the Human-Machine Interface (HMI)?** A) To control and store process data B) To provide a real-time graphical representation for monitoring and controlling machine processes C) To monitor network communication between PLC and sensors D) To convert control signals into physical actions in actuators	**(2Marks)**
04.	**4. Which of the following data security measures should be prioritized when managing sensitive data in an automation system?** A) High-speed data transfer protocols B) Implementation of encryption for data in transit and at rest C) Regular software updates for devices D) Minimizing system downtime for optimal performance	**(Marks)**

05.	**5. When selecting Programmable Logic Controllers (PLCs) for an automation system, which of the following characteristics should be considered to ensure compatibility with the specific application needs?** A) Processing speed and memory capacity B) User interface options and aesthetics C) The brand and model of the communication cables D) Size and weight of the PLC enclosure	**(2Marks)**
06.	**6. What is the primary function of the output modules in a PLC-based automation system?** A) To convert digital signals into analog signals B) To send control signals to actuators and devices in the field C) To monitor the status of sensors in real-time D) To store system configuration and logic programs	**(2Marks)**
07.	**7. In the context of flexible automation, which of the following is the key advantage that allows rapid adaptation to different production processes?** A) High capital investment required B) Automation system is designed for a specific product C) Machines and equipment can be easily reconfigured for different tasks D) Limited human involvement in decision-making	**(2Marks)**
08.	**8. Which of the following is the most suitable type of PLC for complex, multi-stage manufacturing processes that require extensive I/O configurations?** A) Fixed (Compact) PLC B) Modular PLC C) Distributed PLC D) Remote PLC	**(2Marks)**

09.	**9. During the data collection process in an automation system, which technique is most effective for gathering insights directly from system operators and maintenance staff?** A) Surveys B) Observation C) Interviews D) Focus Groups	**(2Marks)**
10.	**10. When configuring data transmission settings for a PLC in an automation system, which of the following configurations is critical for ensuring proper communication with external devices?** A) Device Addressing and Mapping B) Device ID and Access Control C) User Interface Configuration D) File Format Compatibility	**(2Marks)**
11.	The HMI (Human-Machine Interface) is only used for controlling an automation system and does not play a role in monitoring the system's performance. TRUE OR FALSE	**(3Marks)**
12.	In a flexible automation system, the configuration of the machinery and equipment can be easily changed to produce different products with minimal downtime. TRUE OR FALSE	**(3Marks)**
13.	A modular PLC system cannot be expanded or modified once it is installed, as it is designed to be a fixed configuration. TRUE OR FALSE	**(3Marks)**
14.	The communication protocol Modbus RTU is typically used for communication over Ethernet, allowing multiple devices to be connected in a network. TRUE OR FALSE	**(3Marks)**
15.	In an automation system, data from sensors is primarily used for input to the PLC, which then processes the data and sends output signals to actuators and other components. TRUE OR FALSE	**(3Marks)**

16.	**17: Match the types of automation systems with their applications:** ***(10marks)***	**(10Marks)**
	A – Automation System Types / **B – Applications**	
	1. **Fixed Automation** / a. Used for tasks requiring frequent changes in processes or product types.	
	2. **Programmable Automation** / b. Suitable for high-volume, repetitive tasks where the production process is standardized.	
	3. **Flexible Automation** / c. Used for tasks that require automated changes to production based on different product designs.	
	4. **Integrated Automation** / d. Incorporates systems like SCADA, PLCs, and HMIs to manage and optimize manufacturing processes in real-time.	
	5. **Supervisory Control and Data Acquisition (SCADA)** / e. Combines hardware and software components to monitor and control complex industrial processes.	
17.	**18: Match the PLC programming languages with their characteristics:** ***(10marks)***	**(10Marks)**
	A – PLC Programming Languages / **B – Characteristics**	
	1. **Ladder Logic (LAD)** / a. Structured text-based language with high-level programming constructs like loops and conditional statements.	
	2. **Function Block Diagram (FBD)** / b. Graphical programming language that uses function blocks to represent the logic.	

	3. **Structured Text (ST)** 4. **Sequential Function Chart (SFC)** 5. **Instruction List (IL)**	c. Graphical programming language resembling electrical relay logic. d. Ideal for processes with complex sequential operations, uses steps and transitions. e. A low-level textual programming language similar to assembly language.	

Section B: Attempt any three (3) questions **(30 marks)**

18.	Explain the critical advantages and disadvantages of implementing an automation system in industrial applications. How can these factors influence decision-making?	**(10marks)**
19.	**What are the key preparation steps when conducting an automation system site visit, and why is each step crucial?**	**(10marks)**
20.	**Compare and contrast fixed, programmable, flexible, and integrated automation systems in terms of structure, functionality, and application.**	**(10marks)**
21.	**Discuss the importance of drawing standards in automation systems and how they ensure design consistency and operational efficiency.**	**(10marks)**
22.	**What are the essential components of a power and control circuit in an automation system, and how does each function?**	**(10marks)**

Section C: Attempt only one (1) question **(15 marks)**

23.	**Describe the steps involved in assembling and wiring automation system components. Why is each step critical?**	**(15marks)**
24.	**Explain the key factors when selecting and configuring a PLC for an automation project?**	**(15marks)**

END OF ASSESSMENT